TINY RECURSIVE AND HIERARCHICAL REASONING MODELS WITH JEPA AUXILIARY OBJECTIVES

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ABSTRACT. We study whether **Joint Embedding Predictive Architecture (JEPA)** auxiliary objectives improve iterative reasoning models. We evaluate JEPA in conjunction with **Tiny Recursive Models (TRM)** on Sudoku puzzles and observe **limited benefits** due to **dense supervision** and the **task's deterministic structure**. We then extend the analysis to **Hierarchical Reasoning Models (HRM)**, where **intermediate latent abstractions** are explicitly introduced; in this setting, JEPA provides a more principled auxiliary signal by enforcing predictive constraints on unsupervised hierarchical representations. Finally, we analyze the feasibility of incorporating **LLM-JEPA-style objectives into TRM**, discussing both their conceptual appeal and the practical challenges posed by small, structured reasoning domains. Our results highlight that the effectiveness of **predictive representation learning** depends strongly on **task structure**, **supervision density**, and **model architecture**.

1. INTRODUCTION

Recent work has shown that strong reasoning behavior can be achieved not only through scale, but also through architectural inductive biases that enable iterative computation and abstraction. Hierarchical Reasoning Models (HRM) and Tiny Recursive Networks (TRM) are two such approaches, performing **repeated latent refinement** with a learned halting mechanism to dynamically allocate computation. Separately, Joint Embedding Predictive Architectures (JEPA) propose learning abstract representations by predicting masked targets in latent space rather than reconstructing raw inputs. This has shown to lead to models that are more robust and generalizable. This project investigates the interaction between these two ideas. HRMs and TRMs operate in latent space, and so can be naturally extended by incorporating JEPA objectives. Our goal is not only to evaluate whether JEPA improves such models, but to understand *when* and *why* predictive representation learning is effective for iterative and hierarchical reasoning models. We evaluate these models with addition of JEPA on Sudoku puzzles as a benchmark of the models' capacity for generalizable reasoning. Additionally, we implement a preliminary LLM-TRM-JEPA variant on the GSM8K mathematical reasoning dataset to explore whether JEPA-style objectives transfer to natural language reasoning with pretrained representations. We use these results to conclude for which architectural structures and task properties predictive representation learning in the form of JEPA is useful.

1.1. Contributions.

This project makes the following contributions:

- End-to-end implementations of HRM and TRM.
- Extensions for integrating an auxiliary JEPA objective.
- A systematic comparison of TRM, TRM+JEPA, HRM, and HRM+JEPA on Sudoku problems.
- An analysis of the feasibility and limitations of incorporating LLM-JEPA-style objectives into TRM.

2. BACKGROUND

2.1. Sudoku as a Reasoning Benchmark.

Sudoku is a constraint satisfaction problem defined over a 9×9 grid, where each cell must satisfy row, column, and subgrid constraints. Although solutions are fully specified, solving a puzzle requires propagating global constraints across the grid rather than relying on local pattern recognition. This makes Sudoku a controlled benchmark for multi-step reasoning, while its dense and deterministic supervision limits the potential benefits of auxiliary representation learning objectives.

2.2. Iterative Reasoning and Adaptive Computation.

Iterative reasoning models refine internal representations over multiple computation steps, allowing information to be gradually integrated and long-range dependencies to be resolved. This process is closely related to fixed-point iteration and message passing in classical constraint satisfaction algorithms. Adaptive computation further enables models to dynamically allocate computation, halting early on easier instances while continuing refinement on harder ones.

2.3. **Hierarchical Reasoning Models.** Hierarchical Reasoning Models (HRM) explicitly separate reasoning across timescales. High-level latent states capture abstract, slowly varying structure, while low-level states perform rapid, fine-grained updates. Bidirectional interaction between these levels allows global hypotheses to guide local decisions and local evidence to refine abstract representations. Because intermediate latent states are not directly supervised, HRMs provide a natural setting in which auxiliary objectives can regularize abstract structure.

2.4. **Tiny Recursive Models.** Tiny Recursive Models (TRM) perform iterative reasoning through repeated application of a single shared transformation, refining latent and output representations jointly. Parameter sharing encourages convergence toward stable attractors in representation space while reducing model complexity. However, the tight coupling between intermediate states and the supervised output objective constrains the extent to which auxiliary predictive losses can introduce additional learning signals.

2.5. **Predictive Representation Learning with JEPA.** Joint Embedding Predictive Architectures (JEPA) learn representations by predicting masked or missing embeddings rather than reconstructing inputs. By operating directly in latent space, JEPA encourages abstraction and smooth representation geometry. While effective in settings with weak or indirect supervision, its interaction with dense, deterministic reasoning tasks remains less clear.

2.6. **LLM-JEPA.** LLM-JEPA extends the JEPA framework to large language models by applying predictive objectives to high-level semantic representations rather than raw tokens. Typically using asymmetric teacher–student architectures, LLM-JEPA aims to encourage the learning of abstract, predictive latent structures while avoiding token-level reconstruction.

3. Related Work

Joint Embedding Predictive Architectures (JEPA) provide an alternative to reconstruction-based self-supervision by learning abstract representations through prediction in latent space rather than raw input space [1]. By avoiding pixel- or token-level reconstruction, JEPA-style objectives reduce reliance on low-level shortcuts and emphasize semantic structure.

JEPA has since been extended across modalities and temporal scales. HiT-JEPA introduces a hierarchical predictive objective for trajectory representation learning [2], while V-JEPA and V-JEPA 2 scale the framework to video, demonstrating that predictive latent learning can support high-level understanding, prediction, and planning without reconstruction losses [3]. These works highlight the importance of architectural inductive biases when applying JEPA to structured or sequential data.

Separately, Hierarchical Reasoning Models (HRM) and Tiny Recursive Models (TRM) introduce an inductive bias for reasoning through iterative latent refinement with adaptive computation [5, 6]. Rather than increasing model depth, these approaches dynamically allocate computation via explicit halting mechanisms, enabling multi-step reasoning within a compact architecture.

Auxiliary objectives are commonly used to regularize deep models, but their benefit depends on alignment with both the architecture and the primary task. In settings with dense and deterministic supervision, such as Sudoku-style reasoning [7], auxiliary predictive losses may offer limited additional signal.

Our work bridges these lines of research by empirically evaluating how JEPA-style predictive objectives interact with recursive reasoning models. Rather than proposing a new method, we identify the conditions under which predictive representation learning complements, or conflicts with, iterative reasoning.

4. Approach

4.1. **Dataset and Setup.** All models are trained and evaluated on the Sudoku Extreme dataset [7], which contains millions of Sudoku puzzles spanning a wide range of difficulty levels. Due to the large size of the full dataset (3.83 million training data points and 423K test data points) and computational constraints, we train all models on a fixed subset of the training split. We use the provided test split for evaluation and report token-level accuracy and cross-entropy loss. Full-puzzle accuracy is not emphasized in this setting, as it requires near-perfect token accuracy (close to 1.0) to yield meaningful values, which is computationally prohibitive to achieve under our training budget. Unless otherwise specified, no curriculum learning or difficulty-based filtering is applied.

4.2. **Model Variants.** All evaluate the four models, which are trained under comparable optimization settings. Performance is measured using token accuracy and relative changes in cross-entropy loss between the non-JEPA and JEPA variants.

4.2.1. *HRM.* We map input tokens to embeddings via $x = \text{Embed(input)} + \text{PosEmbed}$. The model employs two refinement functions: f_L for low-level token processing and f_H for high-level abstract reasoning, where $n = 81$ is sequence length and d is hidden dimension. Both are implemented as transformer blocks with non-causal self-attention. The hierarchical recursion performs N refinement blocks, each with T low-level updates: $z_L^{(i,t+1)} = f_L(z_L^{(i,t)}, z_H^{(i)}, x)$ for $t = 0, \dots, T-1$ and $i = 0, \dots, N-1$, followed by $z_H^{(i+1)} = f_H(z_H^{(i)}, z_L^{(i,T)})$, starting from learned initial states $z_L^{\text{init}}, z_H^{\text{init}}$. The low-level update combines all information sources as $f_L(z_L, z_H, x) = \text{TransformerBlock}(z_L + z_H + x)$. From a dynamical systems perspective, these updates define an implicit fixed-point iteration converging toward a solution manifold satisfying Sudoku constraints. Self-attention enables global constraint propagation across rows, columns, and subgrids. Final logits are $\hat{y} = W_{\text{out}} z_H^{(N)}$. A halting head predicts $q_t = \sigma(w_h^\top z_H^{(t)}[0])$ to indicate solution completeness. Training uses loss $\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{y}, y) + \lambda_{\text{halt}} \mathcal{L}_{\text{halt}}$ with $\lambda_{\text{halt}} = 0.001$, where $\mathcal{L}_{\text{halt}} = -\sum_{t=1}^{N} y_{\text{halt}} \log q_t + (1 - y_{\text{halt}}) \log(1 - q_t)$. During inference, samples halting when $q_t > 0.5$ exit the batch, enabling adaptive computation allocation.

4.2.2. *HRM+JEPA.* To encourage the learning of predictive and robust latent representations, we augment HRM with a JEPA-style auxiliary objective. At each training iteration, a subset of grid positions is randomly masked, and the model is trained to predict the latent representations of these masked positions using the remaining visible context.

Formally, let H^{ctx} denote latent embeddings produced by the model when masked inputs are provided, and let H^{tgt} denote embeddings produced from the full, unmasked solution. A lightweight predictor network $g(\cdot)$ maps context embeddings to predicted target embeddings. The JEPA loss is defined as:

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|M|} \sum_{i \in M} \left\| g(H_i^{\text{ctx}}) - \text{stopgrad}(H_i^{\text{tgt}}) \right\|_2^2 .$$

Unlike reconstruction-based objectives, JEPA operates directly in representation space. Geometrically, this objective enforces smoothness and linear predictability of latent embeddings: given partial context, the representation of a missing component should lie close to a predictable manifold defined by the visible inputs. This encourages invariance to masking and discourages encoding strategies that rely on brittle, position-specific shortcuts.

In HRM, however, the output representation Y is already directly supervised at every grid position and every recursion step via the token-level cross-entropy loss. As a result, the JEPA objective often duplicates information already enforced by the primary loss, acting primarily as a weak regularizer that smooths early optimization rather than altering the final solution reached by the model.

4.2.3. *TRM.* TRMs assert that the high-level and low-level modules are the same network, decreasing the variance of the model at what has experimentally been little worsening of results. Formally, this corresponds to tying the update functions $f_L = f_H = f$ and applying the same recursion used in HRM,

$$z_L \leftarrow f(z_L + z_H + x), \qquad z_H \leftarrow f(z_L + z_H),$$

with shared parameters across both updates. Like for HRMs, we use non-causal transformer blocks to model the latent updates.

4.2.4. *TRM+JEPA.* JEPA is applied similarly for TRMs as for HRMs.

4.2.5. *LLM-TRM-JEPA.* LLM-TRM-JEPA extends TRM+JEPA to natural language reasoning by replacing learned token embeddings with a frozen pretrained LLM encoder (TinyLlama-1.1B). The LLM processes question–answer pairs from GSM8K to produce contextual embeddings, which are projected into a 64-dimensional TRM reasoning space. A compact TRM module (2.2M trainable parameters) then performs iterative latent refinement via recursive updates and deep supervision. JEPA masking and prediction are applied *post-projection* within the compressed TRM space rather than in the LLM's native embedding dimension.

The model is trained with a composite objective that combines supervised learning, adaptive computation, and predictive representation learning:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{halt}} \mathcal{L}_{\text{halt}} + \lambda_{\text{JEPA}} \mathcal{L}_{\text{JEPA}},$$

where $\mathcal{L}_{\text{CE}}$ is the token-level cross-entropy loss on assistant outputs, $\mathcal{L}_{\text{halt}}$ is a binary cross-entropy loss encouraging early halting on fully correct predictions, and $\mathcal{L}_{\text{JEPA}}$ is a mean-squared error loss between predicted and target latent embeddings at masked positions in the TRM reasoning space.

Unlike HRM+JEPA and TRM+JEPA, where JEPA regularizes representations learned from scratch, LLM-TRM-JEPA applies JEPA to embeddings already grounded in pretrained semantic knowledge. The frozen LLM constrains the input manifold, while the TRM learns task-specific reasoning trajectories, shifting JEPA's role from structure formation to smoothing geometry within a semantically rich latent space.

4.3. **Hyperparameter Selection.** We perform hyperparameter selection separately for each model using a lightweight two-stage search procedure. In the first stage, we run short preview experiments over a grid of model widths, transformer depths, and (where relevant) JEPA loss weights using a small subset of the training data. These preview runs are trained for a limited number of epochs and ranked according to token-level accuracy on the validation set.

In the second stage, the top-performing configuration from each model family is trained for a longer duration on the full training subset. This procedure allows us to efficiently explore the most impactful hyperparameters while avoiding the computational cost of fully training a large number of configurations. All hyperparameter sweeps and selections are implemented using a unified experiment runner, ensuring consistent training and evaluation across the models.

5. Experimental Results

5.1. **Results.** Token accuracy on Sudoku puzzles is measured as the percentage of correctly predicted output tokens at positions that were initially unknown (zero-valued). Across all model families, performance improves steadily with training and converges to stable token-level accuracy after sufficient epochs.

TRM and TRM+JEPA models reach final token accuracy in the high-50% to low-60% range, with higher accuracy achieved by increasing model capacity or transformer depth (Table 1). Across all TRM variants, halt accuracy remains approximately zero, indicating that the learned halting mechanism does not activate reliably under dense token-level supervision.

HRM models achieve comparable final token accuracy to TRM while requiring substantially less computation per epoch. In contrast, HRM+JEPA models exhibit higher preview accuracy during early training but converge to significantly lower final accuracy, indicating reduced convergence quality despite faster initial progress.

Overall, introducing JEPA does not improve final token accuracy under dense supervision. While lightly weighted JEPA objectives are compatible with supervised training, they do not yield consistent gains and can degrade performance in hierarchical reasoning models.

5.2. **Interpretation of Table 1:** Table 1 compares TRM, TRM+JEPA, HRM, and HRM+JEPA under matched training settings, reporting preview token accuracy after 10 epochs, final token accuracy after full training, and computational cost metrics.

For TRM, increasing model capacity via larger hidden size or additional transformer layers consistently improves both preview and final token accuracy, at the cost of increased per-epoch runtime and reduced throughput. In particular, increasing hidden size from 128 to 256 yields the strongest gains, while additional depth provides smaller but consistent improvements.

Adding JEPA to TRM results in similar preview and final accuracy when the auxiliary loss is lightly weighted, indicating that the predictive objective is largely neutral in this regime. Larger JEPA weights reduce final accuracy despite identical architecture and training duration, consistent with over-regularization.

HRM achieves substantially higher throughput and lower epoch times than TRM at comparable hidden sizes, reflecting its shallower recursive structure. While baseline HRM maintains strong final accuracy, HRM+JEPA consistently underperforms at convergence despite higher preview accuracy, suggesting that the auxiliary objective interferes with later-stage hierarchical refinement.

5.3. **Training and Optimization Dynamics.** The trends in Table 1 are reflected in the training curves for TRM, TRM+JEPA, HRM, and HRM+JEPA (Figures 1–5). Across both model families, JEPA primarily affects early optimization, while long-term convergence is governed by the supervised objective and model architecture.

For TRM, adding JEPA results in slightly faster early reductions in cross-entropy loss and marginally higher token accuracy during the first 20–40 epochs. This stabilizing effect is transient: after approximately 50 epochs, the loss and accuracy trajectories of TRM and TRM+JEPA largely overlap, indicating that the auxiliary predictive objective does not alter the final solution under dense supervision.

In contrast, HRM+JEPA exhibits unstable optimization at later stages of training. While early loss decreases mirror those of baseline HRM, the JEPA loss term grows substantially after several hundred epochs and comes to dominate the total loss. This coincides with a sharp rise in training loss and marks a breakdown in optimization when the predictive objective overwhelms supervised signals.

5.4. **Accuracy and Generalization Behavior.** Token accuracy curves (Figures 2 and 3) further highlight the divergence between early and late-stage training. For TRM, training and test token accuracy follow nearly identical trajectories for TRM and TRM+JEPA, with only minor early advantages for the JEPA-augmented model. Final performance and asymptotic error rates are essentially identical, and the generalization gap remains small ($\sim 10^{-3}$) throughout training. For HRM (Figure 4), baseline models reach strong peak token accuracy before gradually declining due to overfitting. HRM+JEPA (Figure 5) attains higher peak accuracy earlier in training but undergoes a substantially larger degradation thereafter, converging to significantly lower training and test accuracy. This degradation coincides with the rise of the JEPA loss term, an increasing generalization gap, and a late-stage increase in token error rate, indicating that the auxiliary objective destabilizes hierarchical refinement rather than improving generalization.

5.5. **HRM, TRM, -JEPA Summary.** Across all experiments, JEPA provides modest benefits during early training by stabilizing optimization and accelerating initial loss reduction. However, under the dense and deterministic supervision of Sudoku, JEPA does not improve final token accuracy or generalization for TRM and substantially degrades convergence for HRM by dominating the optimization objective late in training. These results suggest that predictive auxiliary objectives primarily influence early optimization dynamics, while final performance in dense, fully supervised reasoning tasks is governed by supervised learning signals and architectural inductive biases. In hierarchical models, misalignment between predictive objectives and recursive refinement can lead to instability and degraded generalization.

5.6. **LLM-TRM-JEPA: Empirical Results and Analysis.** We implemented LLM-TRM-JEPA on the GSM8K mathematical reasoning benchmark to evaluate whether JEPA-style predictive objectives provide value when applied to pretrained representations. Training was run for approximately 1,500 batches using TinyLlama-1.1B as a frozen encoder.

Training dynamics. As shown in Figure 7, cross-entropy loss decreases rapidly early in training and continues to improve gradually, stabilizing in the low single-digit range. Total loss follows this downward trend, with residual variance driven by auxiliary components. In contrast, halt loss remains noisy and non-monotonic throughout training, indicating that learning a stable stopping criterion is more challenging than optimizing token-level prediction.

Test-time behavior. Figure 8 shows strong generalization to held-out data. Test cross-entropy loss is low and stable (mean ≈ 1.95), and total loss exhibits limited variance (mean ≈ 2.41). Test halt loss is small (mean ≈ 0.46) and substantially more stable than during training, suggesting that learned halting behavior transfers reasonably well at evaluation time.

Interpretation. Overall, JEPA-augmented TRM training does not interfere with supervised reasoning performance and may implicitly regularize latent dynamics, even when auxiliary losses remain noisy during optimization. The instability of halt loss likely reflects the discrete, instance-dependent nature of adaptive computation rather than representational failure.

6. Discussion

This project demonstrates that the effectiveness of JEPA-style predictive objectives depends strongly on both model architecture and task structure. On Sudoku, the dense and deterministic nature of supervision limits the benefits of predictive representation learning, particularly for TRM-style models, where token-level signals dominate optimization and largely determine final performance. Although hierarchical reasoning models introduce intermediate latent abstractions that are, in principle, more compatible with predictive objectives, our results show that HRM+JEPA primarily affects early training dynamics without improving final accuracy and can destabilize convergence when the auxiliary loss becomes overly influential. In contrast, LLM-TRM-JEPA exhibits a more stable interaction with predictive objectives: when applied on top of frozen LLM representations, JEPA does not degrade supervised performance but also does not yield clear gains in final accuracy, instead influencing latent dynamics and auxiliary behaviors in a manner consistent with a diagnostic or weakly regularizing role. Together, these findings highlight that predictive objectives do not transfer cleanly across domains or architectures, and that architectural intent, representation initialization, and supervision density must be considered jointly when introducing auxiliary losses. Overall, predictive representation learning appears context-dependent; in fully supervised symbolic reasoning tasks such as Sudoku, supervised learning alone is sufficient to induce effective representations, leaving limited room for auxiliary predictive signals. Future work should evaluate larger models on more challenging datasets under increased compute budgets, particularly in settings with sparser supervision and pretrained representations, and explore structured JEPA masking strategies aligned with task constraints, alongside careful comparisons against non-JEPA baselines on held-out tasks to clarify whether JEPA can meaningfully improve generalization in LLM-based reasoning systems.

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APPENDIX A. HYPERPARAMETER-TUNING FOR MODELS

TABLE 1. Comparison of TRM, TRM+JEPA, HRM, and HRM+JEPA configurations on Sudoku. Preview token accuracy is measured after 10 epochs. Final token accuracy is reported after full training (200 epochs) with tuned hyperparameters. Throughput is measured in examples per second. **Note:** For HRM, the "Layers" column denotes the number of layers per module; HRM consists of separate low-level and high-level modules, each with two layers.

Model	Hidden	Layers	JEPA Weight	Preview Acc	Final Acc	Epoch Time	Throughput
TRM	128	1	–	0.12	0.57	**45.1 s**	**222** ex/s
TRM	128	2	–	0.13	0.59	90.0 s	111 ex/s
TRM+JEPA	128	1	5e-5	0.12	0.57	46.0 s	217 ex/s
TRM+JEPA	128	2	5e-5	0.13	0.59	92.0 s	108 ex/s
TRM	256	1	–	**0.14**	**0.61**	**180.0 s**	**55 ex/s**
TRM+JEPA	256	1	1e-4	**0.14**	**0.61**	185.0 s	53 ex/s
TRM+JEPA	256	2	5e-5	0.12	0.58	440.0 s	25 ex/s
HRM	128	2	–	0.15	0.58	**19.9 s**	**200 ex/s**
HRM+JEPA	128	2	5e-6	**0.19**	**0.47**	44.6 s	45 ex/s
HRM	256	2	–	0.16	**0.61**	31.4 s	64 ex/s
HRM+JEPA	256	2	5e-6	**0.19**	0.47	180.0 s	11 ex/s

APPENDIX B. TRM, TRM+JEPA EXPERIMENTAL PLOTS

This appendix contains supplementary plots referenced in the main text, providing additional insight into the training dynamics and generalization behavior of TRM and TRM+JEPA. Both models are trained on a fixed subset of 20,000 Sudoku puzzles and share identical architectural and optimization settings.

Model architecture: Hidden size 256, one TRM layer with four attention heads, two latent refinement steps, and a single refinement block. Deep supervision is applied for a maximum of two supervision steps. JEPA uses a spatial masking ratio of 0.3.

Optimization: Models are trained using AdamW with learning rate 3×10^{-4}, weight decay 0.01, batch size 512, gradient clipping at 1.0, and halt loss weight 1×10^{-3}.

Auxiliary objective: For TRM+JEPA, an additional JEPA loss is applied with weight 1×10^{-4}.

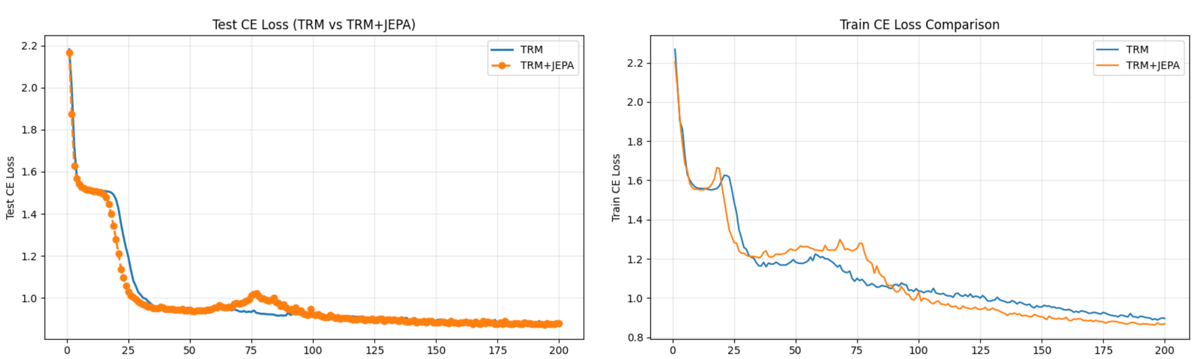

FIGURE 1. Test (left) and training (right) cross-entropy loss for TRM and TRM+JEPA on Sudoku.

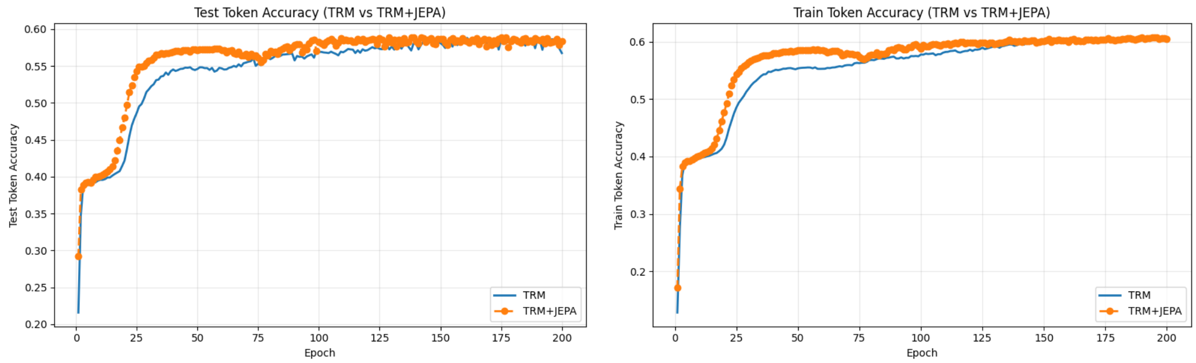

FIGURE 2. Test (left) and training (right) token accuracy for TRM and TRM+JEPA.

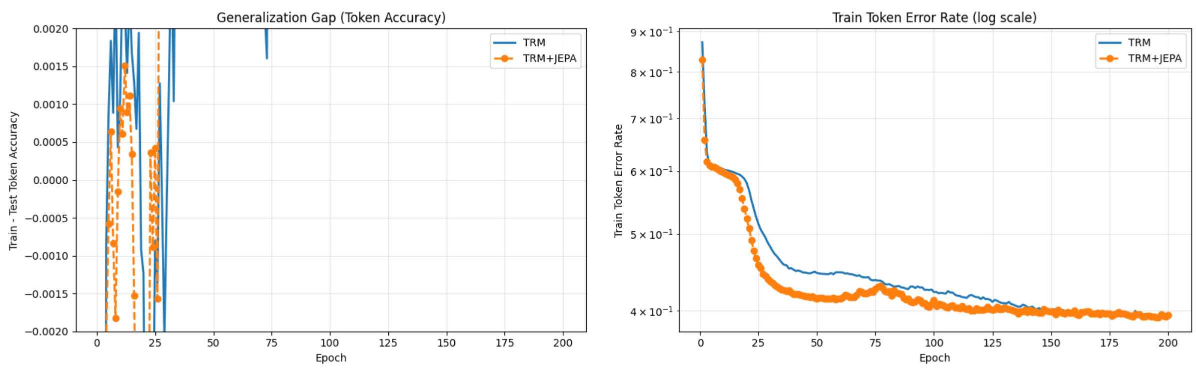

FIGURE 3. Generalization gap (left) and training token error rate on a log scale (right).

APPENDIX C. HRM, HRM+JEPA EXPERIMENTAL PLOTS

This appendix contains supplementary training curves referenced in the main text, providing additional insight into the optimization dynamics of HRM and HRM+JEPA on Sudoku. Both runs are trained on a fixed subset of 5,000 puzzles (Sudoku extreme-mini).

HRM background: HRM belongs to the same family of recursive/iterative refinement models as TRM-style approaches, but it introduces HRM-specific control knobs (refinement blocks, supervision depth, and halting behavior) that do not map 1:1 onto a single "TRM layers" column.

HRM model architecture: Hidden size 256, 2 layers, 4 attention heads, 2 low refinements, 3 refinement blocks, and deep supervision up to 4 supervision steps, with halting probability threshold 0.5 and dropout 0.0.

HRM+JEPA model architecture: Hidden size 128, 2 layers, 4 attention heads, 2 low refinements, 2 refinement blocks, and deep supervision up to 2 supervision steps, with halting probability threshold 0.5 and dropout 0.0. JEPA uses a spatial masking ratio of 0.3 (minimum 8 target tokens) and a 2-layer predictor with hidden size 128, with stop-gradient on target representations.

Optimization: AdamW with learning rate 3×10^{-4}, weight decay 0.01, batch size 128, gradient clipping at 1.0, halt loss weight 1×10^{-3}, trained for 1,000 .

Auxiliary objective: An additional JEPA loss is applied with weight 5×10^{-6}, predicting masked target representations in embedding space from visible context representations.

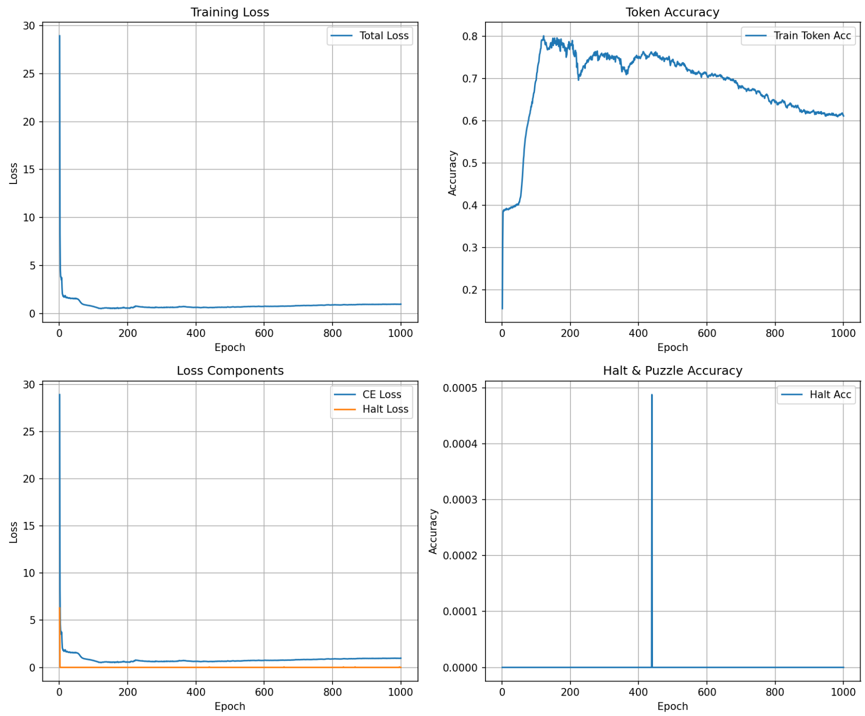

FIGURE 4. HRM training dynamics: total loss, token accuracy, loss components (cross-entropy and halt loss), and halt accuracy over 1,000 epochs.

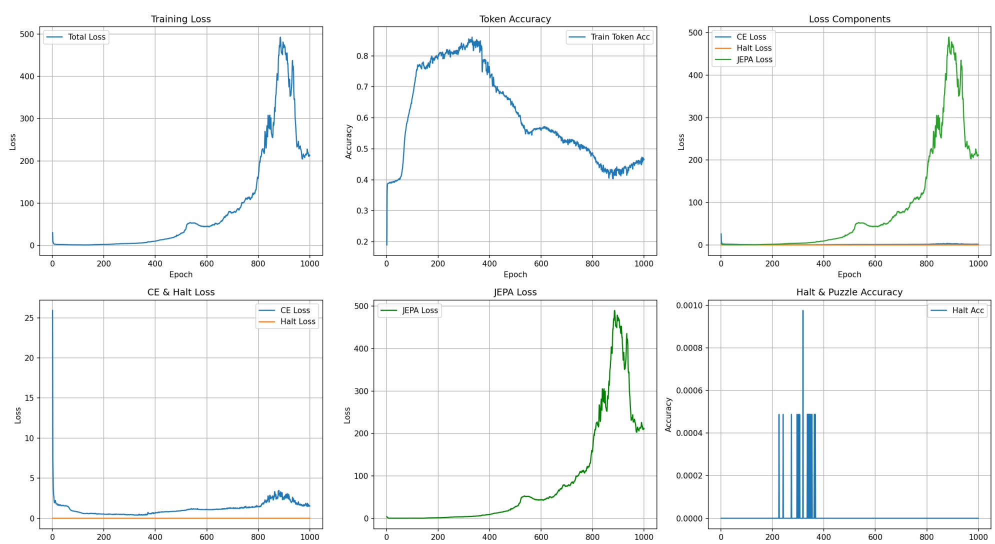

FIGURE 5. HRM+JEPA training dynamics. The JEPA loss term grows substantially late in training and dominates the total loss, coinciding with a degradation in token accuracy.

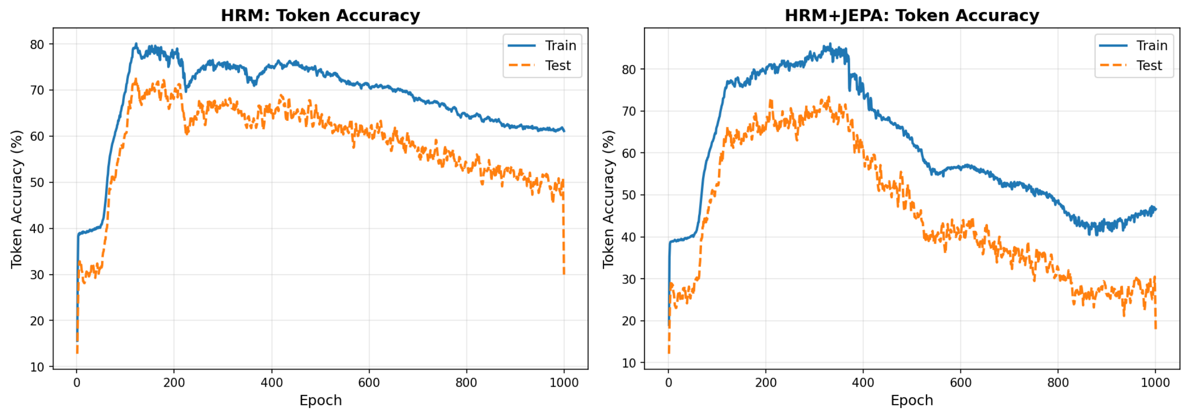

FIGURE 6. Evaluation dynamics for HRM and HRM+JEPA: Training and test token accuracy curves, showing overfitting behavior and the degradation in generalization after approximately 400 epochs.

APPENDIX D. LLM-JEPA TRAINING CURVES (TINYLLAMA ON GSM8K)

This appendix presents the training dynamics of the LLM-JEPA experiment conducted on the GSM8K dataset using a distilled `TinyLlama` architecture. The model employs a Joint Embedding Predictive Architecture (JEPA) auxiliary objective alongside the primary language modeling task. The curves below correspond to batch-level averages across approximately 380 iterations of fine-tuning.

Model configuration: The base model is TinyLlama (1.1B parameters, frozen), with trainable JEPA prediction heads (2.2M parameters) attached to the intermediate embedding space. The JEPA predictor uses a 2-layer MLP with hidden dimension 256 to predict masked token representations from context.

Optimization: AdamW optimizer with learning rate 3×10^{-4}, batch size 4, weight decay 0.01, gradient clipping at 1.0, and JEPA loss weight 1×10^{-4}. Each iteration corresponds to a mini-batch update over mixed-length GSM8K question–answer pairs with 30% spatial masking for JEPA targets.

Interpretation: The total and cross-entropy losses show smooth convergence with typical optimization noise. Without recursive supervision mechanisms, the training trajectory is simpler than TRM-based variants. The JEPA loss increases gradually as the model learns richer representations, making masked prediction targets more challenging and diverse. This serves as a proxy for representation quality and embedding space coverage during training.

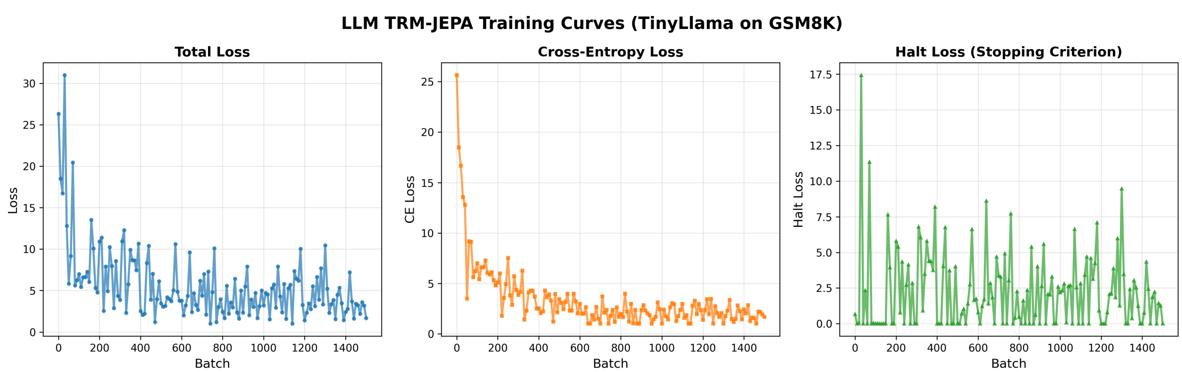

FIGURE 7. Training losses for LLM-JEPA on GSM8K using TinyLlama. Left: Total loss. Center: Cross-entropy loss. Right: Halt loss.

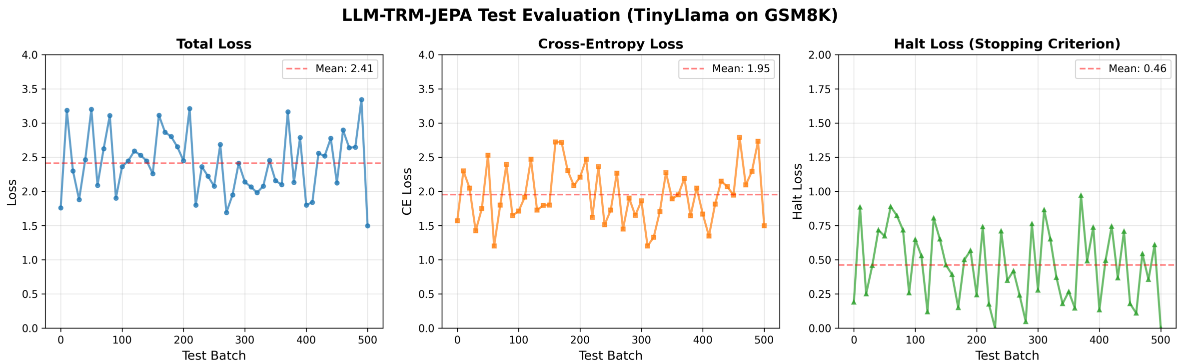

FIGURE 8. Test evaluation losses for LLM-TRM-JEPA on GSM8K using TinyLlama. Evaluated on 500 batches of held-out test data using weights from training batch 520. Dashed red lines indicate mean performance across the test set.